7.2 Graphing Rational Functions

Name _____ Period ____ Date _____

Algebra II 7.2 · Student sheet

I can…

- graph $y = a/(x - h) + k$ and identify its asymptotes, domain, and range.
- find the asymptotes of $y = (ax + b)/(cx + d)$.
- rewrite a rational function to see how it transforms $y = 1/x$.

Vocabulary · fill in the missing word

Rational function — $f(x) = \frac{p(x)}{q(x)}$, where p and q are _____ and $q(x) \neq 0$

Hyperbola — the _____ of $y = \frac{a}{x}$ (two branches)

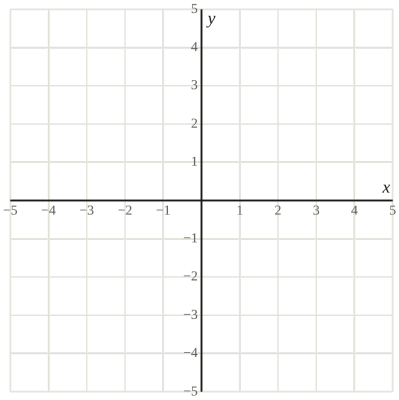
Vertical asymptote — a vertical line $x = h$ the graph _____ but never touches

Horizontal asymptote — a horizontal line $y = k$ the graph _____ as x gets very large or very small

Part 1 · Table to curve

Fill in the table for $y = 1/x$. Plot the points and sketch both branches.

x	−4	−2	−1	−½	−¼	¼	½	1	2	4
y										



Why is there no point at $x = 0$? Does the graph ever reach $y = 0$? _____

Part 2 · Notes: fill in the blanks

$y = a/(x - h) + k$: vertical asymptote _____, horizontal asymptote _____

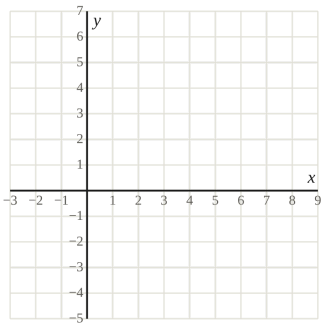
Domain: all real numbers except _____. Range: all real numbers except _____.

$y = (ax + b)/(cx + d)$: vertical asymptote $x =$ _____, horizontal asymptote $y =$ _____

Part 3 · Examples (work them with the class)

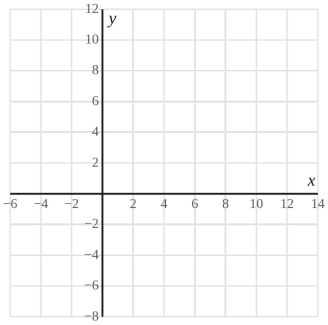
Example 1

Graph $g(x) = \dfrac{2}{x - 3} + 1$. Give the domain and range.



Example 2

Find the asymptotes of $f(x) = \dfrac{2x + 1}{x - 4}$. Give the domain and range.



Example 3

Find the asymptotes of $h(x) = \dfrac{3x - 2}{2x + 6}$.

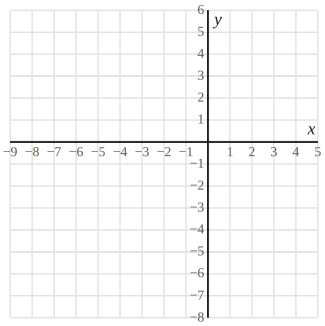
Example 4

Rewrite $g(x) = \dfrac{2x + 5}{x + 1}$ in the form $\dfrac{a}{x - h} + k$. Describe the transformation of $f(x) = \dfrac{1}{x}$.

Part 4 · Partner practice

Partner 1

Graph $y = -\dfrac{4}{x + 2} - 1$. Give the asymptotes, domain, and range.



Partner 2

Find the asymptotes of $y = \dfrac{x - 6}{3x + 3}$.